

Zero-Knowledge proofs

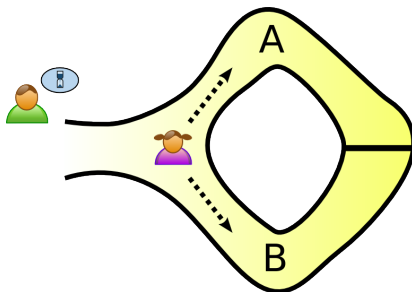
Motivation

- ▶ Secret club, only those can enter who know the password.
- ▶ You can not say the password loud because others may hear it.
- ▶ You convince the guard by answering some question s.t. you only know the answer if yo know the passwords.

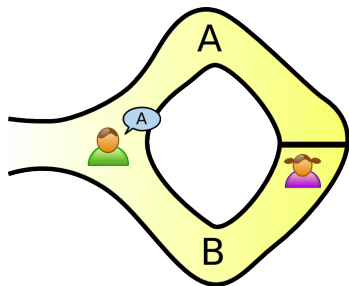
- ▶ Participants: Prover (P), Verifier (V)
- ▶ None of them trust in the other
- ▶ P wants to convince V that some given statement is true without reveal anything more then the mere fact of that statement's truth
- ▶ V wants to make sure whether the statement is really true and expose P if he is lying.

The Ali Baba cave

- ▶ Cave: shaped like a ring, with entrance on one side, magic door blocking the opposite side.
- ▶ Peggy (prover) claims that she has uncovered the secret word to open a magic door in a cave.
- ▶ Victor wants to know whether Peggy knows the secret word.
- ▶ Peggy does not want to reveal her knowledge.

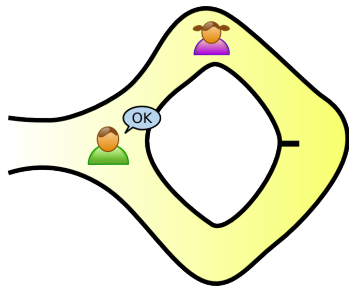


The Ali Baba cave



Protocol
(they run it multiple times):

- ▶ Peggy randomly takes path A or B, while Victor waits outside.
- ▶ Victor chooses an exit path.
- ▶ Peggy appears at the exit Victor names.



Analyze

Honest Peggy can convince Victor:

- ▶ If Peggy does not know the secret word, she can not do better than choosing a random path and hopes Victor chooses the same path.
- ▶ In that case, Peggy is caught with probability 0.5.
- ▶ If they run the protocol multiple times (20), the probability that Peggy does not know the word but gets away with it is $(1/2)^{20} \approx 1/10^6$.

Victor does not learn anything.

- ▶ Victor only sees Peggy come out on one of the paths.

Color blind friend

Story

- ▶ Victor is color-blind.
- ▶ Victor has two balls, a red and the green, but for Victor, the balls are identical.
- ▶ Peggy wants to convince Victor, that the balls are different.

- ▶ Victor puts the two balls behind his back and chooses one of them and shows this to Peggy.
- ▶ Victor puts back the ball behind his back, and chooses one of them and shows it to Peggy and asks: "Did I switch the ball?"
- ▶ Peggy answers honestly.
- ▶ The whole procedure is repeated multiple times.

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Color blind analysis

- ▶ If Peggy is honest (and the two balls are different), she can answer the question right every time.
- ▶ If the two balls are identical, then Peggy can not do better than take a random guess. Peggy answers right with probability $1/2$.
- ▶ If the process is repeated 30 times, the probability that Peggy answers right every time is $\approx 1/10^9$ (which is smaller than winning the lottery).

Zero-Knowledge proof

- ▶ **Completeness:** If the statement is true, then an honest verifier (that is, one following the protocol properly) will be convinced of this fact by an honest prover.
- ▶ **Soundness:** if the statement is false, then no cheating prover can convince an honest verifier that it is true, except with some small probability.
- ▶ **Zero-knowledge:** if the statement is true, then no verifier learns anything other than the fact that the statement is true.

Interaction

- ▶ Interactive: the prover and the verifier can communicate, the verifier can ask questions and the prover answers them.
- ▶ Non-interactive: no interaction between the prover and the verifier. The prover sends a zero-knowledge proof to the verifier, who can accept or decline it.

Fiat-Shamir heuristic

Transforms an interactive ZKP (where the verifier's randomness is public) into a non-interactive ZKP in the random oracle model.

Hamiltonian cycle

Some hard problems

- ▶ Finding a Hamiltonian cycle given a large graph is hard.
- ▶ Deciding if two large graphs are isomorphs is also hard.

Peggy wants to prove to Victor that she knows a Hamiltonian cycle in a large graph G without revealing the cycle itself.

- ▶ P creates a graph H isomorphic to G .
- ▶ V can choose from the following:
 - ▶ P must show the Hamiltonian cycle in H .
 - ▶ P must reveal the isomorphism between G and H .

Hamiltonian cycle

- ▶ If P is honest, she can convince Victor
- ▶ If P does not know the Hamiltonian cycle in G , then she has two options.
 1. Construct H that is not isomorphic to G , but she knows a Hamiltonian cycle in H .
 2. Construct H that is isomorphic to G . In that case P does not know a Hamiltonian cycle in H .
- ▶ No matter how P chooses, she gets caught with probability $1/2$.
 - ▶ If P chooses the first strategy she gets caught if V asks for the isomorphism
 - ▶ If P chooses the second strategy she gets caught if V asks for Hamiltonian cycle
- ▶ If P and V repeat the challenge multiple times, then dishonest P gets caught with almost 1 probability.

Sudoku

Peggy wants to prove to Victor that she can solve a Sudoku puzzle without revealing the solution.

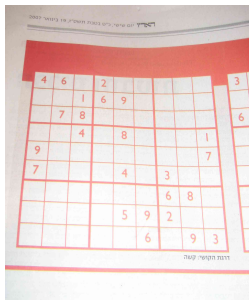
- ▶ P will put cards on a large, 9×9 Sudoku table. V will collect them and check the properties of Sudoku.
- ▶ P makes 27 from each type of card.
- ▶ In the filled-in cells, she places three cards with the assigned value facing up.
- ▶ In the rest of the cells, P places three cards according to the solution, facing down.
- ▶ For each row, column, and subgrid, V chooses one of the three cards of each cell in the corresponding row/column/subgrid and makes packets out of the chosen cards.
- ▶ P shuffles each 27 packets separately.
- ▶ V verifies that all numbers appear in each packet.

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Sudoku



Three stages of the protocol.

- ▶ The Sudoku puzzle.
- ▶ 3 cards in every cell. Face-up cards for the filled-in-cells and face down cards for the rest.
- ▶ Collecting cards: one from each row / column / subgrid

Source: https://www.wisdom.weizmann.ac.il/~naor/PAPERS/SUDOKU_DEMO/

Sudoku

- ▶ Correctness: Clearly, honest P can convince V .
- ▶ Soundness:
 - ▶ If P cannot solve the Sudoku, she has to take some guesses. There will be cells where the three cards are not identical.
 - ▶ One of the choices of these cards will result in P being caught.
 - ▶ Repeat the procedure multiple times to reduce the probability that P cheats and gets away with it.

Remark

If there are at least t cells, where the three cards of P are not identical, then the probability that P is not caught is at most $1/(2/3)^t$.

Discrete log

- ▶ Public values: q prime number and $g, y \in \mathbb{Z}_q^*$.
- ▶ Peggy wants to prove that she knows x such that $g^x = y$.
- ▶ Calculating x from g and y is hard if q is large enough.

Protocol

- ▶ P chooses a random $v \in \mathbb{Z}_q^*$ and sends $t = g^v$ to V .
- ▶ V chooses a random $c \in \mathbb{Z}_q^*$ and sends c to P .
- ▶ P computes $r = v - cx \pmod{q-1}$ and sends r to V .
- ▶ V checks if $t = g^r y^c$.

Discrete log

- ▶ Correctness: $g^r y^c = g^{v-cx} g^{cx} = g^v = t$ because $g^{q-1} \equiv 1 \pmod{q}$.
- ▶ Soundness:
 - ▶ Suppose P doesn't know x but can compute r s.t. $t = g^r y^c$.
 - ▶ Then P can compute the discrete log of g and $t/y^c = g^v / y^c = g^{v-cx}$.
 - ▶ As c is chosen by V after P commits to v , and x is unknown to P it contradicts the hardness of discrete log in $\mathbb{Z}_q^*$.
- ▶ Zero-knowledge: V cannot compute v from t , and x is masked by v in r , so V cannot compute x .

Random oracle

Random oracle (RO)

- ▶ Oracle (black-box) that responds to every unique query with a truly random response chosen uniformly.
 - ▶ If a query is repeated, it responds in the same way every time that query is submitted.
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- ▶ RO is an idealized hash function.
 - ▶ It is used to prove the security of cryptographic systems.
 - ▶ Motivation: If we use a secure hash instead, its probably still secure.
 - ▶ Some made-up schemes are secure in ROM but insecure if any real function is substituted.

Fiat-Shamir heuristic

Suppose Π is an interactive zero-knowledge protocol where V 's random coin tosses are public. Fiat-Shamir transforms it to a non-interactive protocol in the random oracle model.

Non-interactive discrete log

Peggy wants to prove that she knows x such that $y = g^x$ without revealing x .

- ▶ P chooses a random $v \in \mathbb{Z}_q^*$ and sends $t = g^v$ to V .
- ▶ P computes $c = H(g, y, t)$ where H is the random oracle (hash function)
- ▶ P computes $r = v - cx \pmod{q-1}$. The resulting proof is the pair (t, r) .
- ▶ Anyone can use this proof to calculate c and check whether $t = g^r y^c$.

Fiat-Shamir heuristic

Comparison

The difference between the interactive and the non-interactive version is the second step. Instead of V choosing a random c , P computes $c = H(g, y, t)$.

- ▶ Correctness: Same as before
- ▶ Zero-Knowledge: Same as before.
- ▶ Soundness:
 - ▶ As the protocol is non-interactive, P can try multiple values of v (and t) to get the value c she wants.
 - ▶ As H is a random oracle, no matter what v (or t) P chooses, the value c is random.

- ▶ If P can choose $c \neq 0$ independently of t , she can cheat.
- ▶ As the protocol is non-interactive, P can compute c prior to t .

Attack

- ▶ P chooses a random r and a random c .
- ▶ P computes and publishes $t = g^r y^c$

Remark

- ▶ P does not know v , that it is matter as P does not need it.
- ▶ The attack does not work in the random oracle model, because the probability of getting t such that $c = H(g, y, t)$ is negligible.

Commitments

- ▶ Allows one to commit to a chosen value (or chosen statement) while keeping it hidden to others, with the ability to reveal the committed value later.
- ▶ Visual:
 - ▶ Commitment phase: The sender places a message in a locked box and gives the box to the receiver, so the sender cannot change its value.
 - ▶ Reveal phase: The sender gives the key of the lock box to the receiver, who can verify the contents.

Coin flipping commitment

- ▶ Alice and Bob want to play a coin flip online.
- ▶ Idea: Alice commits to a value and reveals it after the coin toss.

Coin toss commitment

- ▶ Alice chooses $b \in \{0, 1\}$ (outcome of coin flip), and a random $r \in \{0, 1\}^{128}$.
- ▶ Alice computes $c = H(b, r)$ and sends it to Bob.
- ▶ Bob flips the coin and tells Alice the result.
- ▶ Alice reveals b and r .

If H is a secure hash function, then Alice cannot cheat.

Secret sharing with unreliable dealer

Shamir secret sharing

- ▶ n participants $(P_1, \dots, P_n)$, at least k of them together can recover the secret.
 - ▶ Setup: $\mathbb{F}$ finite field, s.t. $|\mathbb{F}| > n$.
 - ▶ Dealer: chooses a random polynomial p of degree at most $k - 1$. The secret share of party P_i is $p(i)$.
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- ▶ Suppose that the dealer is not trustworthy.
 - ▶ It is impossible to detect if the dealer does not follow the algorithm above to generate the shares.
 - ▶ IF the dealer sends a random share to all participants $\Rightarrow$ every k would reconstruct a different value.

Verifiable secret sharing

The dealer uses commitments to prove the consistency of the shares.

- ▶ Let g be a primitive root in $\mathbb{F}^*$.
- ▶ Dealer publishes $h_i = g^{s_i} = g^{p(i)}$ for $i \in \{1, \dots, n\}$.
- ▶ The parties can compute the approximation in the exponent of g : $h = \prod_{i \in S} h_i^{\lambda_i} = \prod_{i \in S} g^{\lambda_i p(i)} = g^s$ (λ_i is the coefficient in the Lagrange interpolation, that is, $\sum_{i \in S} \lambda_i p(i) = p(0) = s$).